

Set 1

p. 398–400: 34,36,48,52,58,60,76,85–90,95,96,103

p. 418: 84,86,90,92

1. If

$$f(x) = \int_0^{g(x)} \frac{dt}{\sqrt{1+t^3}} \quad \text{and} \quad g(x) = \int_0^{\cos x} [1 + \sin(t^2)] dt,$$

determine the value of $f'(\pi/2)$.

2. Show that the function

$$f(x) := \int_1^x \sqrt{1+t^2} dt$$

is one-to-one. Compute $(f^{-1})'(0)$.

The next pair of questions is taken from *Basic Analysis: Japanese Grade 11*, translated and published by the American Mathematical Society.

3. Find a (continuous) function f such that

$$f(x) = x + \int_0^2 f(t) dt$$

for every real number x .

4. Find a function f and a number a such that

$$\int_1^x f(t) dt = x^3 + ax - 5$$

for every real number x .